

Student Depression Data Analysis Project

2025-10-29

Research questions:

1. Is there a correlation between the cumulative grade point average of the student and depression?
2. Is age associated with depression?
3. Is academic pressure associated with depression?
4. Are students with higher work/study hours more likely to experience depression?
5. Do dietary habits (e.g., healthy vs unhealthy) relate to depression?
6. Does financial stress correlate with depression?

Loading data and initial data exploration

```
df <- read.csv("./dataset/student_depression_dataset.csv")
```

```
head(df)
```

```
##   id Gender Age      City Profession Academic.Pressure Work.Pressure CGPA
## 1  2  Male  33 Visakhapatnam   Student              5              0 8.97
## 2  8  Female 24      Bangalore   Student              2              0 5.90
## 3 26  Male  31      Srinagar   Student              3              0 7.03
## 4 30  Female 28      Varanasi   Student              3              0 5.59
## 5 32  Female 25      Jaipur    Student              4              0 8.13
## 6 33  Male  29      Pune      Student              2              0 5.70
##   Study.Satisfaction Job.Satisfaction Sleep.Duration Dietary.Habits
## 1                  2                0      '5-6 hours'      Healthy
## 2                  5                0      '5-6 hours'      Moderate
## 3                  5                0 'Less than 5 hours'      Healthy
## 4                  2                0      '7-8 hours'      Moderate
## 5                  3                0      '5-6 hours'      Moderate
## 6                  3                0 'Less than 5 hours'      Healthy
##   Degree Have.you.ever.had.suicidal.thoughts.. Work.Study.Hours
## 1 B.Pharm                      Yes                3
## 2   BSc                      No                3
## 3    BA                      No                9
## 4   BCA                      Yes                4
## 5 M.Tech                      Yes                1
## 6   PhD                      No                4
##   Financial.Stress Family.History.of.Mental.Illness Depression
## 1                1.0                      No                1
## 2                2.0                      Yes                0
## 3                1.0                      Yes                0
```

```
## 4          5.0          Yes          1
## 5          1.0          No           0
## 6          1.0          No           0
```

```
names(df)
```

```
## [1] "id"
## [2] "Gender"
## [3] "Age"
## [4] "City"
## [5] "Profession"
## [6] "Academic.Pressure"
## [7] "Work.Pressure"
## [8] "CGPA"
## [9] "Study.Satisfaction"
## [10] "Job.Satisfaction"
## [11] "Sleep.Duration"
## [12] "Dietary.Habits"
## [13] "Degree"
## [14] "Have.you.ever.had.suicidal.thoughts.."
## [15] "Work.Study.Hours"
## [16] "Financial.Stress"
## [17] "Family.History.of.Mental.Illness"
## [18] "Depression"
```

The data set consists of 27901 observations and 18 features

```
dim(df)
```

```
## [1] 27901    18
```

```
unique(df$Profession)
```

```
## [1] "Student"           "'Civil Engineer'"
## [3] "Architect"         "'UX/UI Designer'"
## [5] "'Digital Marketer'" "'Content Writer'"
## [7] "'Educational Consultant'" "Teacher"
## [9] "Manager"           "Chef"
## [11] "Doctor"            "Lawyer"
## [13] "Entrepreneur"      "Pharmacist"
```

Convert to data.table

```
dt <- as.data.table(df)
```

```
dt[Profession != "Student"]
```

```
##      id Gender  Age      City      Profession
##      <int> <char> <num>    <char>      <char>
## 1:   609  Male   21  Ahmedabad  'Civil Engineer'
## 2:  2105 Female  31   Vadodara    Architect
```

##	3:	9483	Female	32	Kalyan	'UX/UI Designer'
##	4:	11470	Female	28	Ahmedabad	'Digital Marketer'
##	5:	12425	Male	25	Kalyan	'Content Writer'
##	6:	16196	Male	33	Jaipur	Architect
##	7:	16959	Female	24	Mumbai	Architect
##	8:	18291	Male	32	Agra	'Educational Consultant'
##	9:	19226	Male	31	Rajkot	Teacher
##	10:	25193	Male	24	Kalyan	Teacher
##	11:	43035	Male	23	Patna	Manager
##	12:	61219	Female	23	Kalyan	Chef
##	13:	62410	Female	34	Ghaziabad	'Content Writer'
##	14:	72541	Female	28	Hyderabad	Architect
##	15:	78083	Male	32	Meerut	Doctor
##	16:	81442	Male	23	Kanpur	Lawyer
##	17:	88879	Female	29	Agra	Chef
##	18:	89052	Female	29	Meerut	Doctor
##	19:	92226	Male	26	Bhopal	Architect
##	20:	104123	Male	28	Ludhiana	'Digital Marketer'
##	21:	105237	Female	21	Chennai	Entrepreneur
##	22:	105595	Male	28	Jaipur	Teacher
##	23:	111939	Male	24	Bangalore	Architect
##	24:	121474	Male	27	Kalyan	Teacher
##	25:	123259	Male	23	Kanpur	Teacher
##	26:	127070	Male	32	Meerut	Pharmacist
##	27:	127637	Female	25	Delhi	Architect
##	28:	129499	Female	21	Kalyan	'Digital Marketer'
##	29:	132550	Male	25	Visakhapatnam	Teacher
##	30:	136642	Male	29	Ludhiana	Pharmacist
##	31:	140522	Female	22	Vasai-Virar	Architect
##		id	Gender	Age	City	Profession
##		Academic.Pressure	Work.Pressure	CGPA	Study.Satisfaction	Job.Satisfaction
##		<num>	<num>	<num>	<num>	<num>
##	1:	5	0	7.83	1	0
##	2:	5	0	6.95	3	0
##	3:	3	0	9.60	2	0
##	4:	5	0	9.72	3	0
##	5:	5	0	8.50	2	0
##	6:	2	0	7.13	4	0
##	7:	3	0	8.89	5	0
##	8:	3	0	5.74	4	0
##	9:	3	0	7.48	5	0
##	10:	4	0	9.66	5	0
##	11:	3	0	8.62	2	0
##	12:	3	0	9.91	4	0
##	13:	5	0	6.83	4	0
##	14:	3	0	6.99	3	0
##	15:	5	0	9.70	3	0
##	16:	4	0	6.88	1	0
##	17:	2	0	6.86	3	0
##	18:	3	0	6.35	3	0
##	19:	4	0	7.28	3	0
##	20:	1	0	9.41	5	0
##	21:	3	0	8.17	5	0
##	22:	3	0	5.82	4	0

## 23:	5	0	7.09	4	0
## 24:	3	0	5.89	3	0
## 25:	3	0	5.59	1	0
## 26:	4	0	8.54	1	0
## 27:	5	0	8.98	3	0
## 28:	5	0	6.91	4	0
## 29:	5	0	7.25	1	0
## 30:	4	0	5.77	4	0
## 31:	3	0	9.13	4	0
##	Academic.Pressure	Work.Pressure	CGPA	Study.Satisfaction	Job.Satisfaction
##	Sleep.Duration	Dietary.Habits	Degree		
##	<char>	<char>	<char>		
## 1:	'5-6 hours'	Healthy	MSc		
## 2:	'Less than 5 hours'	Moderate	BSc		
## 3:	'7-8 hours'	Moderate	PhD		
## 4:	'More than 8 hours'	Healthy	MA		
## 5:	'5-6 hours'	Moderate	B.Ed		
## 6:	'More than 8 hours'	Moderate	MSc		
## 7:	'More than 8 hours'	Unhealthy	MSc		
## 8:	'7-8 hours'	Healthy	B.Ed		
## 9:	'5-6 hours'	Unhealthy	MD		
## 10:	'7-8 hours'	Moderate	BE		
## 11:	'More than 8 hours'	Healthy	BHM		
## 12:	'7-8 hours'	Healthy	BHM		
## 13:	'7-8 hours'	Moderate	MA		
## 14:	'5-6 hours'	Moderate	BSc		
## 15:	'Less than 5 hours'	Healthy	M.Pharm		
## 16:	'More than 8 hours'	Unhealthy	LLM		
## 17:	'7-8 hours'	Moderate	BHM		
## 18:	'Less than 5 hours'	Healthy	MD		
## 19:	'More than 8 hours'	Healthy	BSc		
## 20:	'5-6 hours'	Healthy	BA		
## 21:	'Less than 5 hours'	Moderate	MCA		
## 22:	'5-6 hours'	Moderate	MD		
## 23:	'Less than 5 hours'	Unhealthy	BBA		
## 24:	'Less than 5 hours'	Unhealthy	MSc		
## 25:	'More than 8 hours'	Healthy	BCA		
## 26:	'7-8 hours'	Unhealthy	MBBS		
## 27:	'More than 8 hours'	Healthy	B.Arch		
## 28:	'Less than 5 hours'	Moderate	M.Com		
## 29:	'7-8 hours'	Unhealthy	B.Ed		
## 30:	'More than 8 hours'	Healthy	B.Pharm		
## 31:	'7-8 hours'	Unhealthy	MCA		
##	Sleep.Duration	Dietary.Habits	Degree		
##	Have.you.ever.had.suicidal.thoughts...	Work.Study.Hours	Financial.Stress		
##		<char>	<num>	<char>	
## 1:		Yes	11	3.0	
## 2:		No	8	1.0	
## 3:		Yes	5	3.0	
## 4:		Yes	10	5.0	
## 5:		Yes	0	2.0	
## 6:		No	2	5.0	
## 7:		Yes	3	5.0	
## 8:		Yes	3	5.0	

## 9:	Yes	12	2.0
## 10:	Yes	12	3.0
## 11:	No	9	5.0
## 12:	Yes	10	3.0
## 13:	Yes	5	5.0
## 14:	Yes	10	5.0
## 15:	Yes	3	4.0
## 16:	Yes	2	4.0
## 17:	Yes	12	4.0
## 18:	No	12	1.0
## 19:	No	12	4.0
## 20:	Yes	5	3.0
## 21:	Yes	4	3.0
## 22:	Yes	5	1.0
## 23:	Yes	1	3.0
## 24:	Yes	2	5.0
## 25:	Yes	11	4.0
## 26:	Yes	2	5.0
## 27:	Yes	5	5.0
## 28:	No	10	5.0
## 29:	Yes	1	1.0
## 30:	Yes	4	2.0
## 31:	Yes	4	1.0
##	Have.you.ever.had.suicidal.thoughts.. Work.Study.Hours Financial.Stress		
##	Family.History.of.Mental.Illness Depression		
##	<char>	<int>	
## 1:	Yes	1	
## 2:	Yes	1	
## 3:	No	1	
## 4:	Yes	1	
## 5:	Yes	1	
## 6:	Yes	1	
## 7:	Yes	1	
## 8:	Yes	1	
## 9:	Yes	1	
## 10:	No	1	
## 11:	No	1	
## 12:	No	1	
## 13:	No	1	
## 14:	Yes	1	
## 15:	No	1	
## 16:	No	1	
## 17:	Yes	1	
## 18:	No	1	
## 19:	Yes	0	
## 20:	Yes	0	
## 21:	No	1	
## 22:	No	0	
## 23:	Yes	1	
## 24:	No	1	
## 25:	Yes	1	
## 26:	Yes	1	
## 27:	No	1	
## 28:	No	1	

```
## 29:                Yes          1
## 30:                No           1
## 31:                Yes          1
##   Family.History.of.Mental.Illness Depression
```

Count observations by profession

```
dt[, .N, by="Profession"]
```

```
##           Profession      N
##           <char> <int>
##  1:           Student 27870
##  2:      'Civil Engineer'    1
##  3:           Architect    8
##  4:      'UX/UI Designer'    1
##  5:      'Digital Marketer'    3
##  6:      'Content Writer'    2
##  7: 'Educational Consultant'    1
##  8:           Teacher    6
##  9:           Manager    1
## 10:           Chef    2
## 11:           Doctor    2
## 12:           Lawyer    1
## 13:      Entrepreneur    1
## 14:           Pharmacist    2
```

Count observations by city There are observations of students from 52 cities.

```
dt[, .N, by="City"][order(-N)]
```

```
##           City      N
##           <char> <int>
##  1:           Kalyan 1570
##  2:           Srinagar 1372
##  3:           Hyderabad 1340
##  4:      Vasai-Virar 1290
##  5:           Lucknow 1155
##  6:           Thane 1139
##  7:           Ludhiana 1111
##  8:           Agra 1094
##  9:           Surat 1078
## 10:           Kolkata 1066
## 11:           Jaipur 1036
## 12:           Patna 1007
## 13:      Visakhapatnam  969
## 14:           Pune  968
## 15:      Ahmedabad  951
## 16:           Bhopal  934
## 17:           Chennai  885
## 18:           Meerut  825
## 19:           Rajkot  816
## 20:           Delhi  768
```

```
## 21:      Bangalore 767
## 22:      Ghaziabad 745
## 23:      Mumbai 699
## 24:      Vadodara 694
## 25:      Varanasi 685
## 26:      Nagpur 651
## 27:      Indore 643
## 28:      Kanpur 609
## 29:      Nashik 547
## 30:      Faridabad 461
## 31:      Saanvi 2
## 32:      Bhavna 2
## 33:      City 2
## 34:      Harsha 2
## 35:      M.Tech 1
## 36:      'Less Delhi' 1
## 37:      3.0 1
## 38: 'Less than 5 Kalyan' 1
## 39:      Mira 1
## 40:      Vaanya 1
## 41:      Gaurav 1
## 42:      Harsh 1
## 43:      Reyansh 1
## 44:      Kibara 1
## 45:      Rashi 1
## 46:      ME 1
## 47:      M.Com 1
## 48:      Nalyan 1
## 49:      Mihir 1
## 50:      Nalini 1
## 51:      Nandini 1
## 52:      Khaziabad 1
##      City N
```

Explore categories of “Sleep.Duration”

```
unique(dt$Sleep.Duration)
```

```
## [1] "'5-6 hours'" "'Less than 5 hours'" "'7-8 hours'"
## [4] "'More than 8 hours'" "Others"
```

Explore categories of “Dietary.Habits”

```
unique(dt$Dietary.Habits)
```

```
## [1] "Healthy" "Moderate" "Unhealthy" "Others"
```

Explore categories of “Degree”

```
unique(dt$Degree)
```

```
## [1] "B.Pharm"      "BSc"          "BA"           "BCA"          "M.Tech"
## [6] "PhD"          "'Class 12'"    "B.Ed"         "LLB"          "BE"
## [11] "M.Ed"         "MSc"          "BHM"          "M.Pharm"      "MCA"
## [16] "MA"           "B.Com"        "MD"           "MBA"          "MBBS"
## [21] "M.Com"        "B.Arch"       "LLM"          "B.Tech"       "BBA"
## [26] "ME"           "MHM"          "Others"
```

Categories of "Have.you.ever.had.suicidal.thoughts."

```
unique(dt$Have.you.ever.had.suicidal.thoughts..)
```

```
## [1] "Yes" "No"
```

Categories of "Financial.Stress"

```
unique(dt$Financial.Stress)
```

```
## [1] "1.0" "2.0" "5.0" "3.0" "4.0" "?"
```

Categories of "Family.History.of.Mental.Illness"

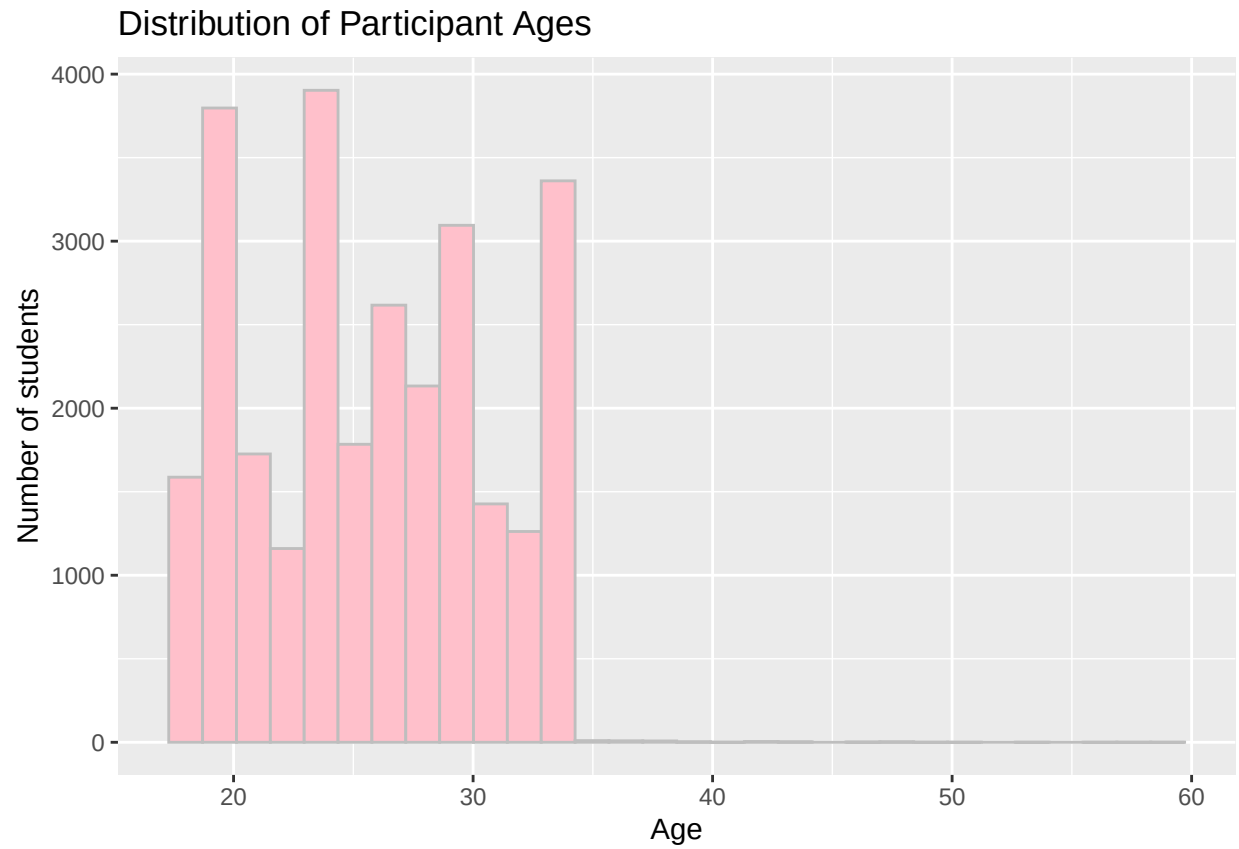
```
unique(dt$Family.History.of.Mental.Illness)
```

```
## [1] "No" "Yes"
```

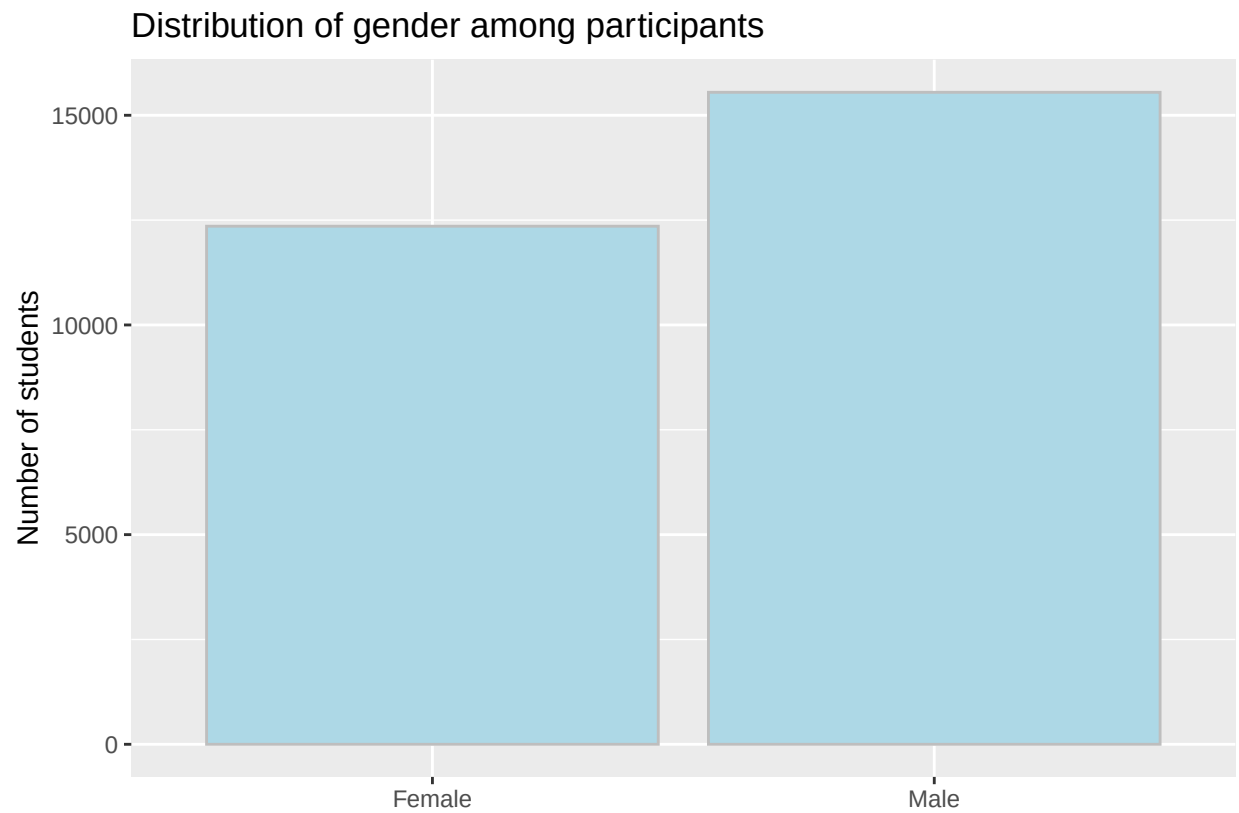
Distribution of age

```
ggplot(dt, aes(x = Age)) +
  geom_histogram(fill="pink", color="grey") +
  ylab("Number of students") +
  xlab("Age") +
  ggtitle("Distribution of Participant Ages")
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```

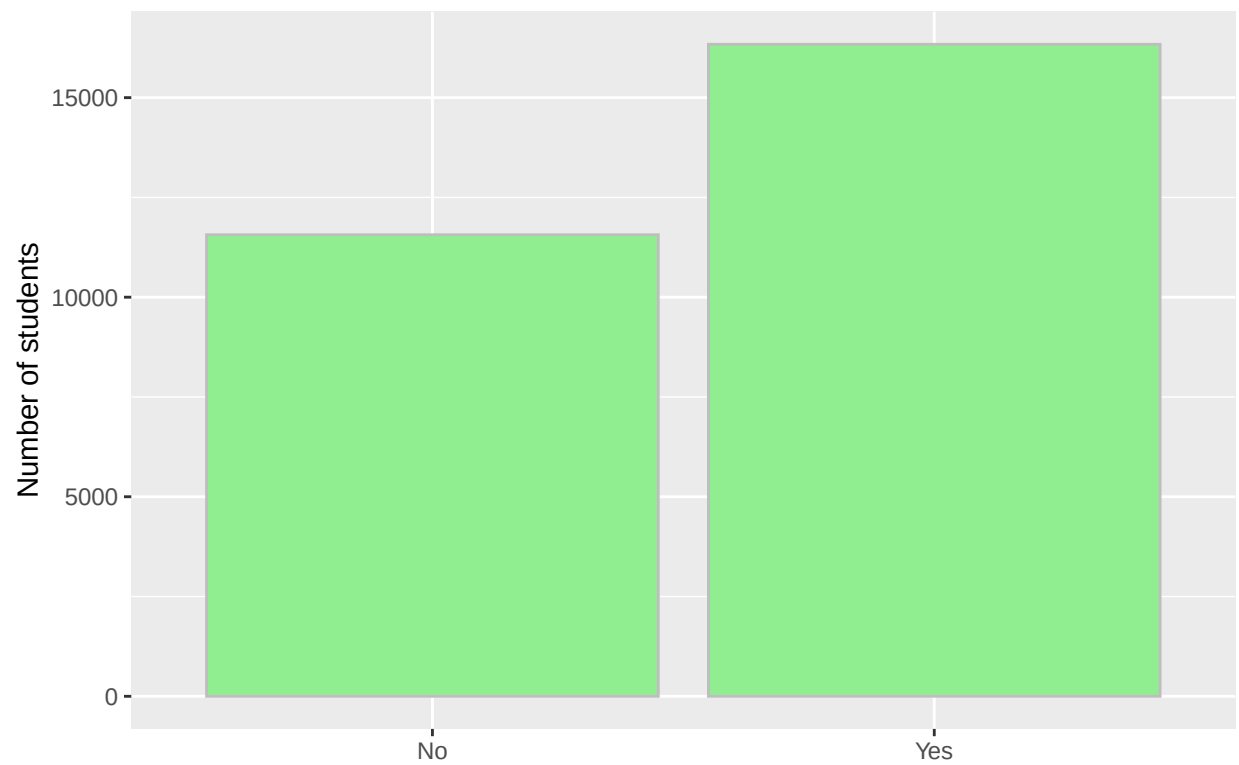



```
ggplot(dt, aes(x = Gender)) +  
  geom_bar(fill="lightblue", color="grey") +  
  ylab("Number of students") +  
  xlab("") +  
  ggtitle("Distribution of gender among participants")
```



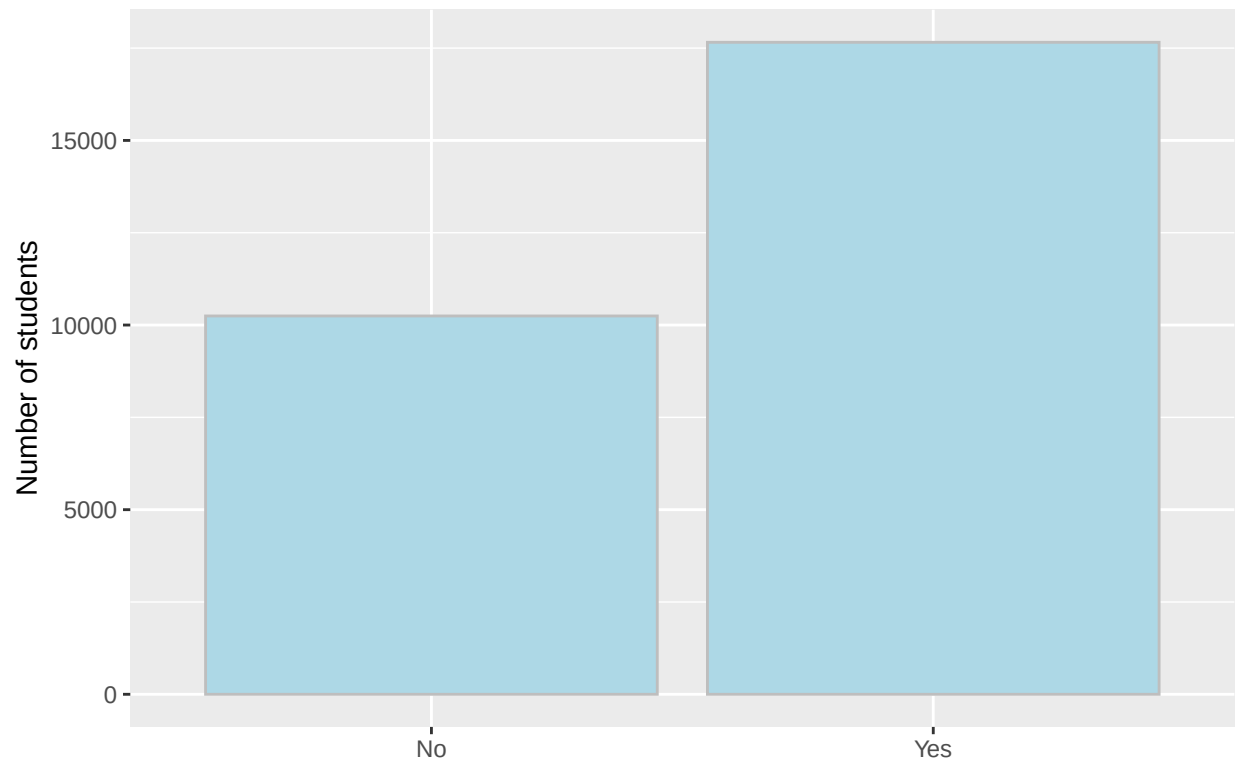
```
ggplot(dt, aes(x = factor(Depression, levels=c(0, 1), labels=c("No", "Yes")))) +  
  geom_bar(fill="lightgreen", color="grey") +  
  ylab("Number of students") +  
  xlab("") +  
  ggtitle("Distribution of depressive vs. non-depressive participants")
```

Distribution of depressive vs. non-depressive participants



```
ggplot(dt, aes(x = Have.you.ever.had.suicidal.thoughts..)) +  
  geom_bar(fill="lightblue", color="grey") +  
  ylab("Number of students") +  
  xlab("") +  
  ggtitle("Distribution of students with/without suicidal thoughts")
```

Distribution of students with/without suicidal thoughts



Data preprocessing

`head(dt)`

```
##      id Gender  Age      City Profession Academic.Pressure Work.Pressure
##    <int> <char> <num>    <char>    <char>          <num>          <num>
## 1:     2  Male   33 Visakhapatnam Student             5             0
## 2:     8 Female  24  Bangalore Student             2             0
## 3:    26  Male   31  Srinagar Student             3             0
## 4:    30 Female  28  Varanasi Student             3             0
## 5:    32 Female  25   Jaipur Student             4             0
## 6:    33  Male   29    Pune Student             2             0
##      CGPA Study.Satisfaction Job.Satisfaction Sleep.Duration Dietary.Habits
##    <num>          <num>          <num>          <char>          <char>
## 1:  8.97             2             0      '5-6 hours'      Healthy
## 2:  5.90             5             0      '5-6 hours'      Moderate
## 3:  7.03             5             0 'Less than 5 hours'      Healthy
## 4:  5.59             2             0      '7-8 hours'      Moderate
## 5:  8.13             3             0      '5-6 hours'      Moderate
## 6:  5.70             3             0 'Less than 5 hours'      Healthy
##      Degree Have.you.ever.had.suicidal.thoughts.. Work.Study.Hours
##    <char>          <char>          <num>
## 1: B.Pharm             Yes             3
## 2:   BSc              No             3
## 3:   BA              No             9
## 4:   BCA             Yes             4
## 5: M.Tech             Yes             1
```

```
## 6:      PhD                                     No                                     4
##      Financial.Stress Family.History.of.Mental.Illness Depression
##              <char>                                <char>      <int>
## 1:              1.0                                     No          1
## 2:              2.0                                     Yes          0
## 3:              1.0                                     Yes          0
## 4:              5.0                                     Yes          1
## 5:              1.0                                     No          0
## 6:              1.0                                     No          0
```

Select variables of interest

```
dt <- dt[, .(id, Age, Gender, Academic.Pressure, CGPA, Study.Satisfaction, Sleep.Duration, Dietary.Habits)]
```

Map boolean values to numerical

```
dt[, Have.you.ever.had.suicidal.thoughts.. := ifelse(Have.you.ever.had.suicidal.thoughts.. == "Yes", 1L, 0L)]
```

```
dt[, Family.History.of.Mental.Illness := ifelse(Family.History.of.Mental.Illness == "Yes", 1L, 0L)]
```

```
dt$Gender <- as.factor(dt$Gender)
dt$Sleep.Duration <- as.factor(dt$Sleep.Duration)
dt$Dietary.Habits <- as.factor(dt$Dietary.Habits)
dt$Financial.Stress <- as.numeric(dt$Financial.Stress)
```

Warning: NAs introduced by coercion

```
setnames(dt, old="Family.History.of.Mental.Illness", new="Family.Mental.Illness")
setnames(dt, old="Have.you.ever.had.suicidal.thoughts..", new="Suicidal.thoughts")
```

```
head(dt)
```

```
##      id  Age Gender Academic.Pressure  CGPA Study.Satisfaction
##      <int> <num> <fctr>                <num> <num>                <num>
## 1:      2   33  Male                5  8.97                    2
## 2:      8   24 Female                2  5.90                    5
## 3:     26   31  Male                3  7.03                    5
## 4:     30   28 Female                3  5.59                    2
## 5:     32   25 Female                4  8.13                    3
## 6:     33   29  Male                2  5.70                    3
##      Sleep.Duration Dietary.Habits Suicidal.thoughts Work.Study.Hours
##              <fctr>      <fctr>                <int>      <num>
## 1:      '5-6 hours'    Healthy                1          3
## 2:      '5-6 hours'    Moderate               0          3
## 3: 'Less than 5 hours'    Healthy                0          9
## 4:      '7-8 hours'    Moderate               1          4
## 5:      '5-6 hours'    Moderate               1          1
## 6: 'Less than 5 hours'    Healthy                0          4
##      Financial.Stress Family.Mental.Illness Depression
##              <num>                <int>      <int>
## 1:              1                0          1
```

```
## 2:          2          1          0
## 3:          1          1          0
## 4:          5          1          1
## 5:          1          0          0
## 6:          1          0          0
```

Check for missing values

```
colSums(is.na(df))
```

```
##          id          Gender
##          0          0
##          Age          City
##          0          0
##          Profession Academic.Pressure
##          0          0
##          Work.Pressure CGPA
##          0          0
##          Study.Satisfaction Job.Satisfaction
##          0          0
##          Sleep.Duration Dietary.Habits
##          0          0
##          Degree Have.you.ever.had.suicidal.thoughts..
##          0          0
##          Work.Study.Hours Financial.Stress
##          0          0
##          Family.History.of.Mental.Illness Depression
##          0          0
```

```
str(dt)
```

```
## Classes 'data.table' and 'data.frame': 27901 obs. of 13 variables:
## $ id : int 2 8 26 30 32 33 52 56 59 62 ...
## $ Age : num 33 24 31 28 25 29 30 30 28 31 ...
## $ Gender : Factor w/ 2 levels "Female","Male": 2 1 2 1 1 2 2 1 2 2 ...
## $ Academic.Pressure : num 5 2 3 3 4 2 3 2 3 2 ...
## $ CGPA : num 8.97 5.9 7.03 5.59 8.13 5.7 9.54 8.04 9.79 8.38 ...
## $ Study.Satisfaction : num 2 5 5 2 3 3 4 4 1 3 ...
## $ Sleep.Duration : Factor w/ 5 levels "'5-6 hours'",...: 1 1 3 2 1 3 2 3 2 3 ...
## $ Dietary.Habits : Factor w/ 4 levels "Healthy","Moderate",...: 1 2 1 2 2 1 1 4 2 2 ...
## $ Suicidal.thoughts : int 1 0 0 1 1 0 0 0 1 1 ...
## $ Work.Study.Hours : num 3 3 9 4 1 4 1 0 12 2 ...
## $ Financial.Stress : num 1 2 1 5 1 1 2 1 3 5 ...
## $ Family.Mental.Illness: int 0 1 1 1 0 0 0 1 0 0 ...
## $ Depression : int 1 0 0 1 0 0 0 0 1 1 ...
## - attr(*, ".internal.selfref")=<externalptr>
```

```
summary(dt)
```

```
##          id          Age          Gender          Academic.Pressure
## Min. : 2 Min. :18.00 Female:12354 Min. :0.000
## 1st Qu.: 35039 1st Qu.:21.00 Male :15547 1st Qu.:2.000
```

```
## Median : 70684      Median :25.00              Median :3.000
## Mean   : 70442      Mean   :25.82              Mean   :3.141
## 3rd Qu.:105818     3rd Qu.:30.00              3rd Qu.:4.000
## Max.   :140699     Max.   :59.00              Max.   :5.000
##
##      CGPA      Study.Satisfaction      Sleep.Duration
## Min.   : 0.000   Min.   :0.000      '5-6 hours'      :6183
## 1st Qu.: 6.290   1st Qu.:2.000      '7-8 hours'      :7346
## Median : 7.770   Median :3.000      'Less than 5 hours':8310
## Mean   : 7.656   Mean   :2.944      'More than 8 hours':6044
## 3rd Qu.: 8.920   3rd Qu.:4.000      Others           : 18
## Max.   :10.000   Max.   :5.000
##
##      Dietary.Habits  Suicidal.thoughts  Work.Study.Hours  Financial.Stress
## Healthy : 7651      Min.   :0.0000      Min.   : 0.000   Min.   :1.00
## Moderate : 9921     1st Qu.:0.0000     1st Qu.: 4.000   1st Qu.:2.00
## Others   : 12       Median :1.0000     Median : 8.000   Median :3.00
## Unhealthy:10317     Mean   :0.6328     Mean   : 7.157   Mean   :3.14
##                                     3rd Qu.:1.0000     3rd Qu.:10.000   3rd Qu.:4.00
##                                     Max.   :1.0000     Max.   :12.000   Max.   :5.00
##                                     NA's   :3
## Family.Mental.Illness  Depression
## Min.   :0.000          Min.   :0.0000
## 1st Qu.:0.000          1st Qu.:0.0000
## Median :0.000          Median :1.0000
## Mean   :0.484          Mean   :0.5855
## 3rd Qu.:1.000          3rd Qu.:1.0000
## Max.   :1.000          Max.   :1.0000
##
```

Answer research questions

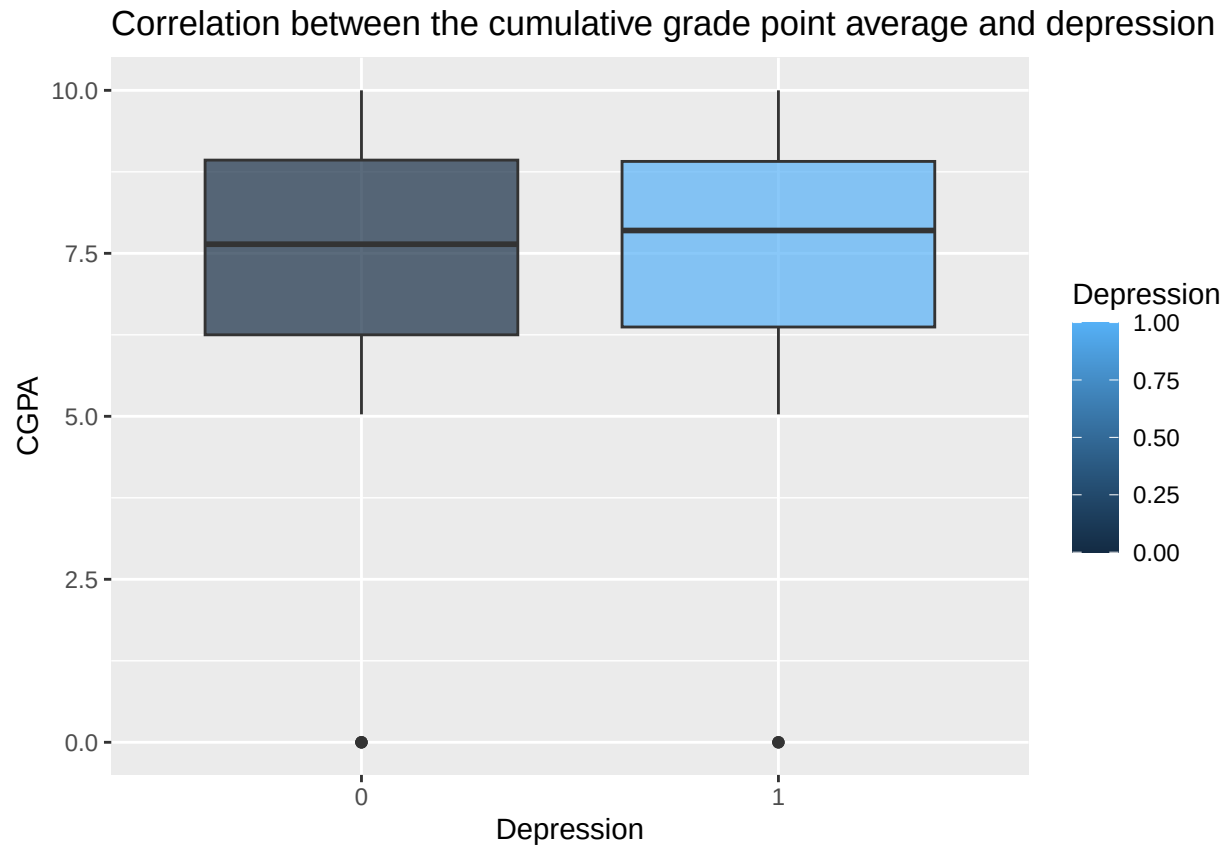
Is there a correlation between the cumulative grade point average of the student and depression?

The following analysis illustrates that there is no statistically significant correlation between those two features.

```
cor(dt$CGPA, dt$Depression)
```

```
## [1] 0.02221047
```

```
ggplot(dt, aes(x=factor(Depression), y=CGPA, fill=Depression)) +
  geom_boxplot(alpha=0.7) +
  labs(
    title="Correlation between the cumulative grade point average and depression",
    x="Depression",
    y="CGPA"
  )
```



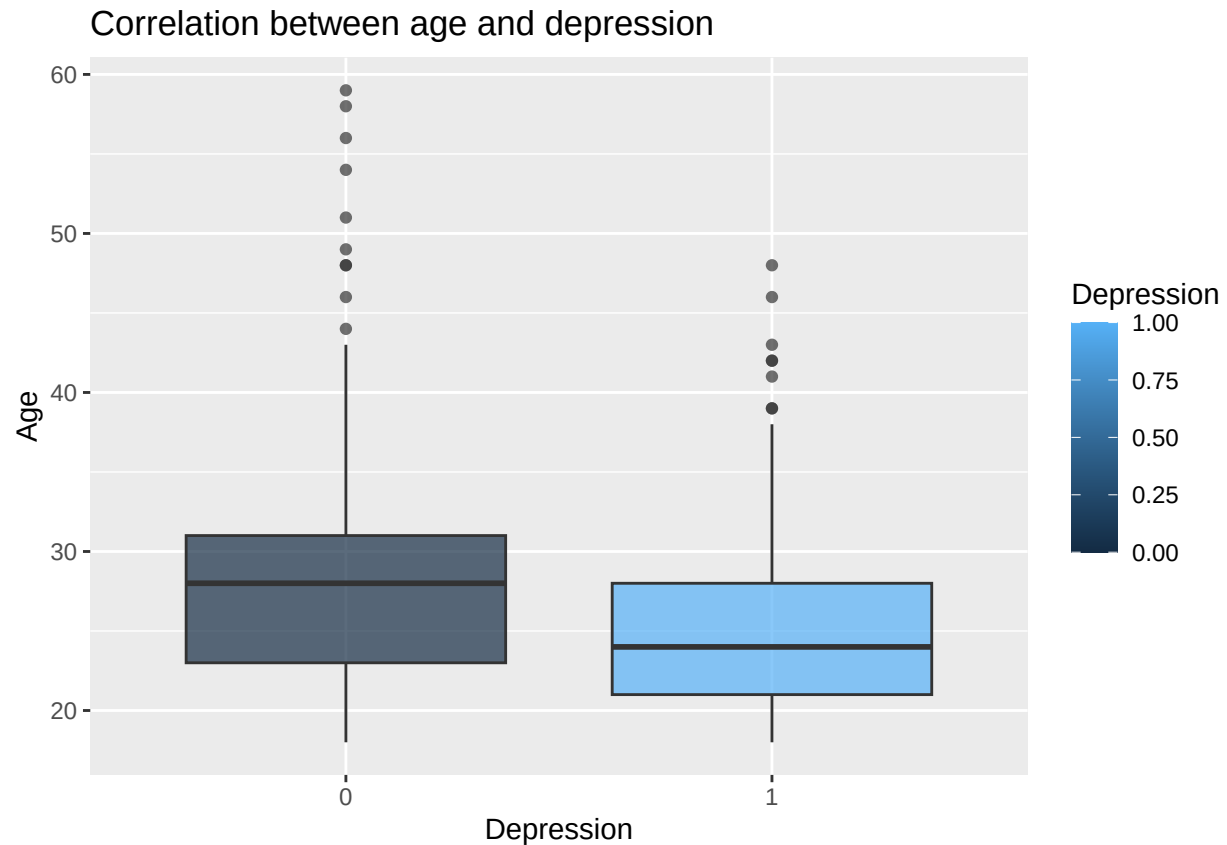
Is age associated with depression?

There is a significant negative correlation between age and depression (-0.22), indicating that younger participants are slightly more likely to experience depression.

```
cor(dt$Age, dt$Depression)
```

```
## [1] -0.226422
```

```
ggplot(dt, aes(x=factor(Depression), y=Age, fill=Depression)) +
  geom_boxplot(alpha=0.7) +
  labs(
    title="Correlation between age and depression",
    x="Depression",
    y="Age"
  )
```

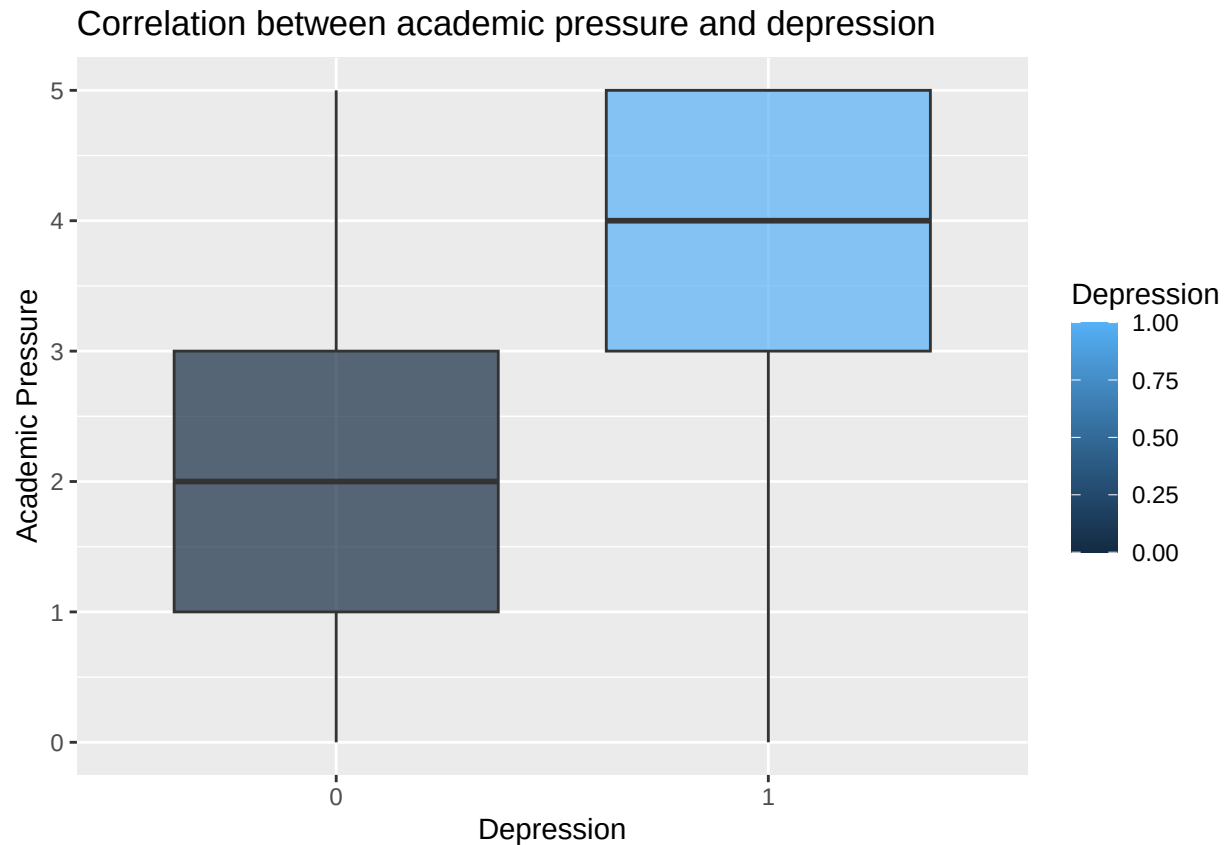
Is academic pressure associated with depression?

There is a very strong correlation between academic pressure and depression (0.47), indicating that students experiencing academic pressure are more likely to experience depression.

```
cor(dt$Academic.Pressure, dt$Depression)
```

```
## [1] 0.4748349
```

```
ggplot(dt, aes(x=factor(Depression), y=Academic.Pressure, fill=Depression)) +  
  geom_boxplot(alpha=0.7) +  
  labs(  
    title="Correlation between academic pressure and depression",  
    x="Depression",  
    y="Academic Pressure"  
  )
```



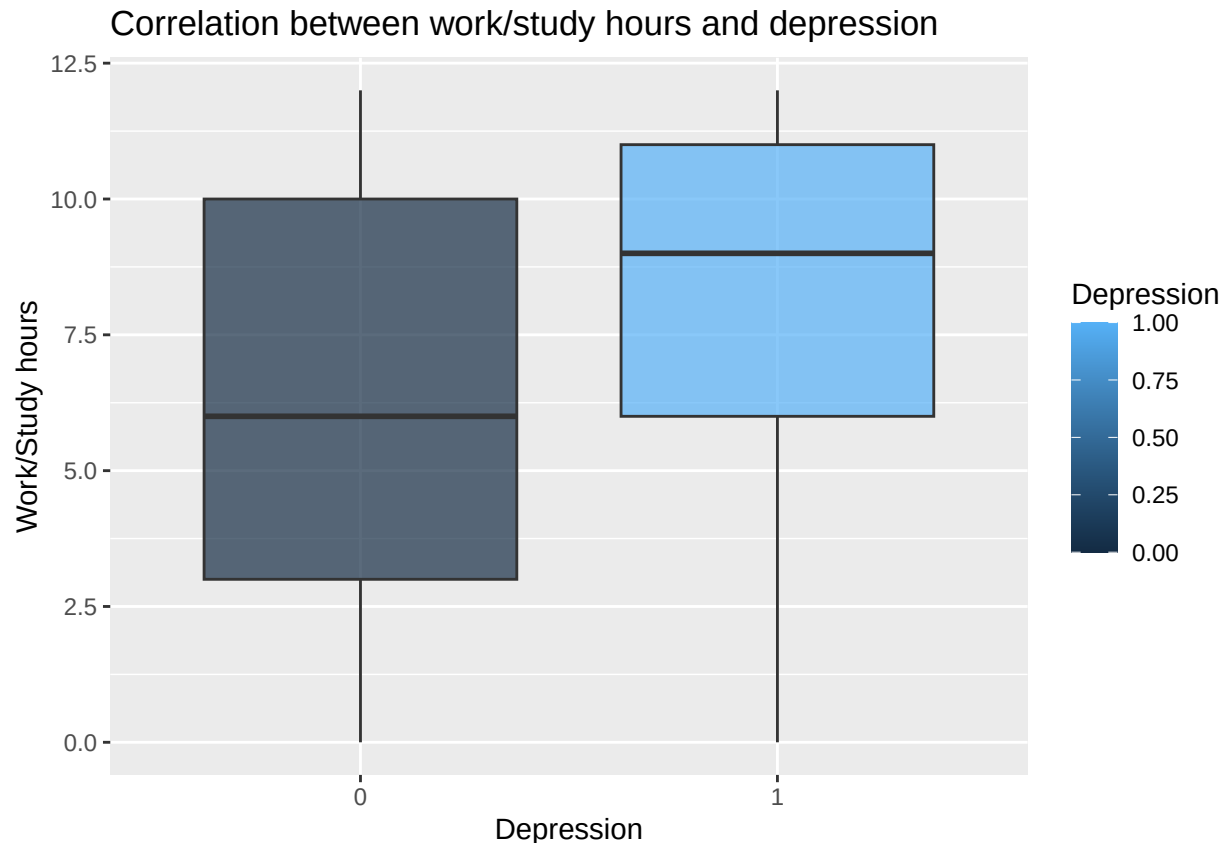
Are students with higher work/study hours more likely to experience depression?

There is a correlation between work/study hours and depression (0.21), indicating that students working and studying more are more likely to experience depression.

```
cor(dt$Work.Study.Hours, dt$Depression)
```

```
## [1] 0.2085628
```

```
ggplot(dt, aes(x=factor(Depression), y=Work.Study.Hours, fill=Depression)) +
  geom_boxplot(alpha=0.7) +
  labs(
    title="Correlation between work/study hours and depression",
    x="Depression",
    y="Work/Study hours"
  )
```



Do dietary habits (e.g., healthy vs unhealthy) relate to depression?

The ANOVA results indicate a significant relationship between dietary habits and depression ($p < 2e-16$). The results are illustrated in the bar plot below.

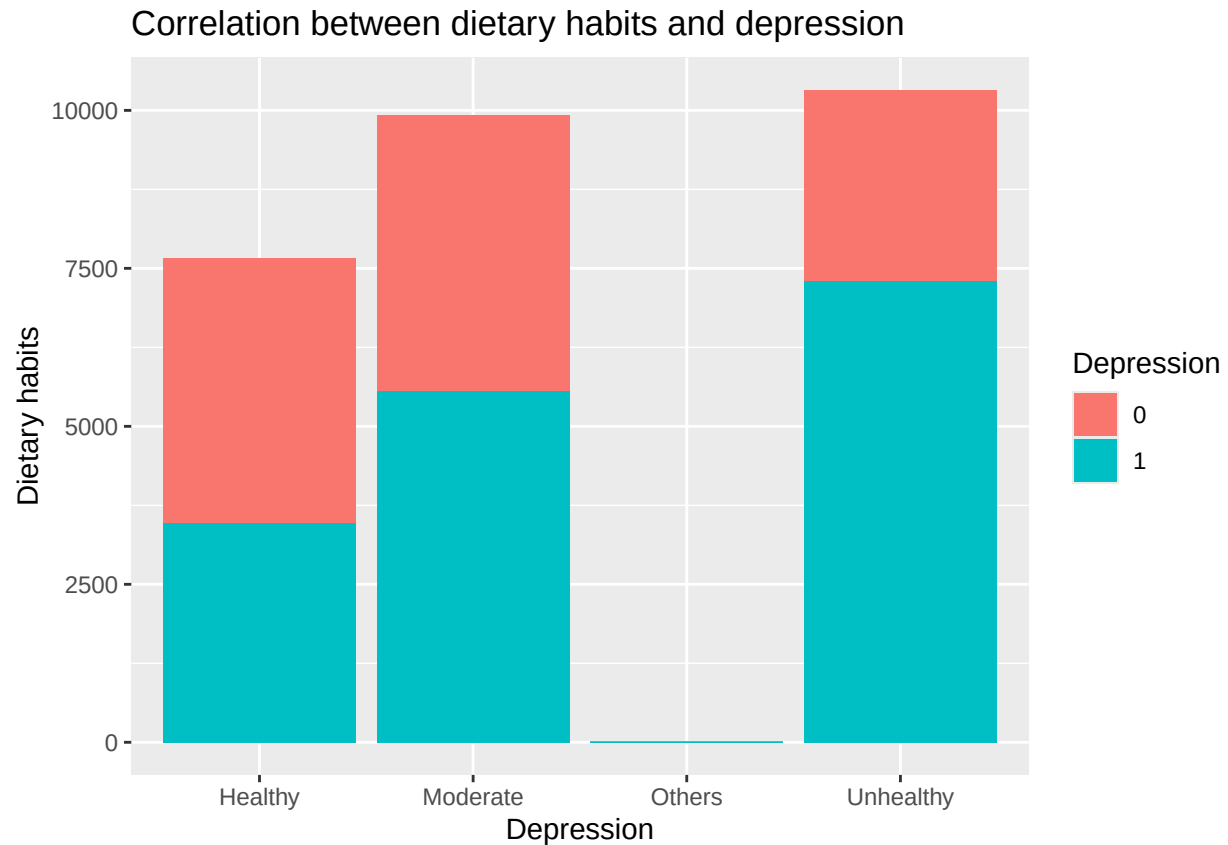
Perform ANOVA-test to examine the correlation

```
anova_result <- aov(as.numeric(Depression) ~ as.factor(Dietary.Habits), data = dt)
```

```
summary(anova_result)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## as.factor(Dietary.Habits)      3      292    97.29   418.9 <2e-16 ***
## Residuals                   27897     6479     0.23
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ggplot(dt, aes(x=Dietary.Habits, fill=as.factor(Depression))) +
  geom_bar(position = "stack") +
  labs(
    title="Correlation between dietary habits and depression",
    x="Depression",
    y="Dietary habits",
    fill="Depression"
  )
```



Does financial stress correlate with depression?

There is a correlation between financial stress and depression (0.36), indicating that students experiencing financial stress are more likely to experience depression.

```
cor(as.numeric(dt$Financial.Stress), as.numeric(dt$Depression))
```

```
## [1] NA
```

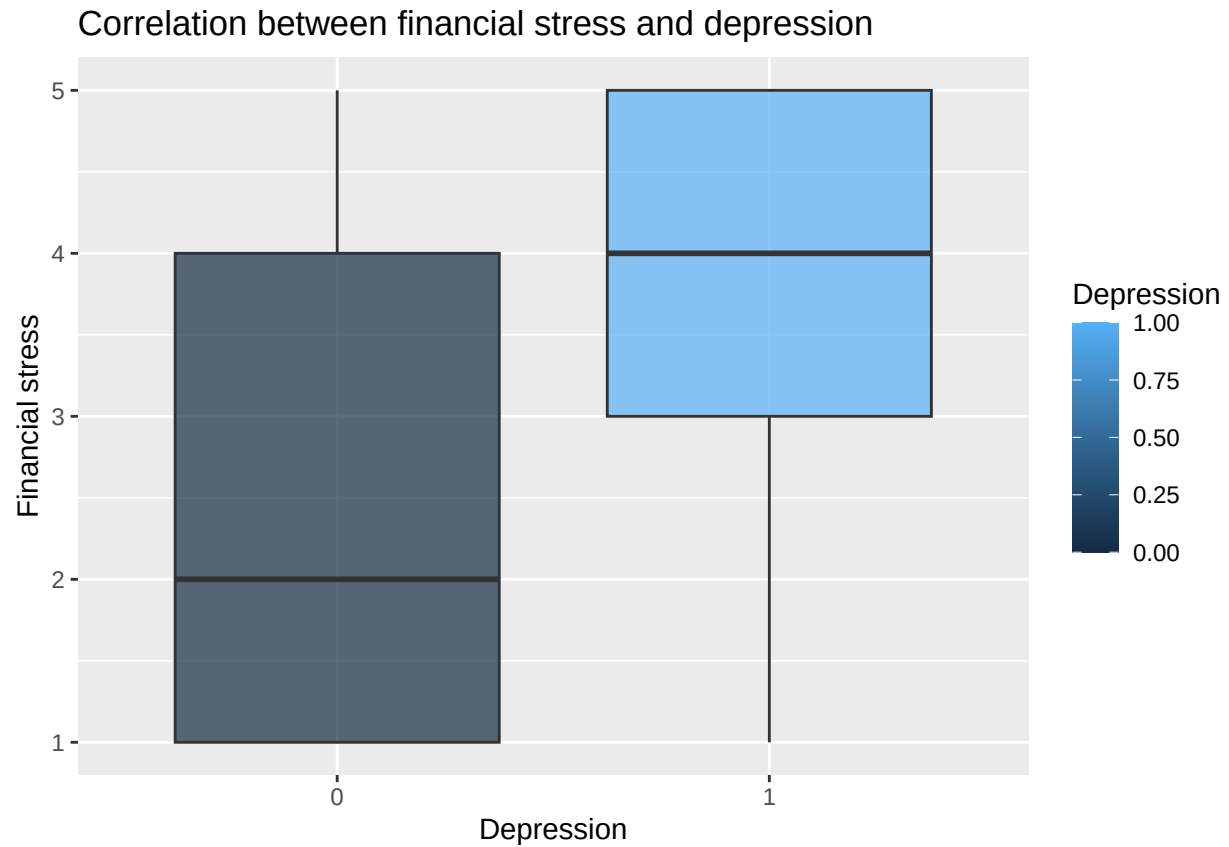
remove NA values

```
dt <- dt[!is.na(Financial.Stress)]
```

```
cor(as.numeric(dt$Financial.Stress), as.numeric(dt$Depression))
```

```
## [1] 0.3635914
```

```
ggplot(dt, aes(x=factor(Depression), y=Financial.Stress, fill=Depression)) +
  geom_boxplot(alpha=0.7) +
  labs(
    title="Correlation between financial stress and depression",
    x="Depression",
    y="Financial stress"
  )
```



Summary

The analysis showed that age, academic and financial pressure, work/study hours, and dietary habits are significantly associated with depression. No significant correlation was found between CGPA and depression.